

8-QUEENS

```
N = int(input("Enter the number of queens: "))
```

```
x = [-1] * N
```

```
def safe(r, c):
```

```
    for i in range(c):
```

```
        if x[i] == r or abs(x[i] - r) == abs(i - c):
```

```
            return False
```

```
    return True
```

```
def queen(c):
```

```
    if c == N:
```

```
        print("\nSolution:")
```

```
        for i in range(N):
```

```
            print(x[i], end=" ")
```

```
        return True
```

```
    for r in range(N):
```

```
        if safe(r, c):
```

```
            x[c] = r
```

```
            if queen(c + 1):
```

```
                return True
```

```
    return False
```

```
if not queen(0):
```

```
    print("No Solution")
```

OUTPUT:

```
Enter the number of queens: 4
```

```
Solution:
```

```
1 3 0 2
```